MIL-M-36954C
12 June 1975
SUPERSEDING
MIL-M-36954B
6 January 1974

MILITARY SPECIFICATION

MASK, SURGICAL, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers disposable lightweight surgical masks, suitable for use by medical personnel in operating rooms and other areas where it is necessary to maintain protective asceptic techniques.

1.2 Classification. The masks shall be of the following types, as specified in the invitation for bids or request for proposals.

Type I - Pleated
Type II - Pouch
Type III - Molded

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-566	Box, Folding, Paperboard.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner

FSC 6515

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SPECIFICATIONS (Cont'd.)

Federal

PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.
PPP-T-76	Tape, Pressure-sensitive Adhesive Paper, Water-Resistant, (For Carton Sealing).

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-vaporproof or Waterproof, Flexible.
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

JOURNAL OF BACTERIOLOGY

Volume 83 Journal of Bacteriology

(Application for copies should be addressed to the Williams and Wilkins Company, Baltimore Md. 21202.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

Munsell Color Company Color Samples (glossy finished).

(Application for samples should be addressed to Munsell Color Company, Inc., 2441 North Calvert Street, Baltimore, Md. 21218).

U. S. PHARMACOPEIAL CONVENTION PUBLICATION
Pharmacopeia of the United States (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Co., Easton, Pa.)

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U.S. DEPARTMENT OF COMMERCE COMMERCIAL STANDARDS
Commercial Standard (CS) 191-53 Flammability of Clothing Textiles

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, Washington, D.C.).

3. REQUIREMENTS

3.1 Material. Except as specified herein, material shall conform to the applicable portions of the specifications and standards listed, and shall be capable of meeting the requirements shown therein. When materials are not specifically designated, those used shall be of the highest commercial quality, and entirely suitable for the purpose intended.

3.1.1 Type I and Type II mask.

3.1.1.1 Facing material. The facing material shall be a non-woven fabric of suitable fibers. The material shall be colorless, or white, or blue, or green and shall be odorless.

3.1.1.2 Filter media. The filter media shall consist of a filter mat composed of pure flame-blown or fine spun glass fibers, or polypropylene fibers and/or other suitable fiber.

3.1.1.3 Tie tapes. The tie tapes shall be made of woven cotton material or constructed from nonwoven fabric. The tie tapes shall be constructed by sewing or shall be impregnated with a thermoplastic, water-resistant adhesive. The tie strings shall finish 3/16 or 7/16 inch wide, and shall not show fraying or loose threads along the cut edges.

3.1.1.3.1 Color. The color of the tape shall be white, or dark blue or dyed to match the color of the mask.

3.1.1.4 Nosepiece. The nosepiece shall be fabricated from a single strip of malleable aluminum, rounded on both ends. The strip shall be 3-1/4 to 4-1/2 inches long and 1/8 to 1/4 inch wide. The thickness shall be 0.017 $\pm$ 0.002 inch. When the nosepiece is to be placed on the inside of the mask, it may be fabricated as described above or from flat, or flattened, malleable, aluminum strips measuring 0.125 $\pm$ 0.005 inch wide and .017 $\pm$ 0.002 inch thick, to .029 $\pm$ .005 inch thick; and extending the length of the mask. The strip shall be malleable enough to be readily bent, and to maintain the contours of the wearer's nose.

3.1.1.4.1 Color. The color of the nosepiece shall be blue, blue-green, green or white. When the nosepiece is placed inside the mask it may be the natural color of the metal.

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3.1.2 Type III mask..

3.1.2.1 Fabric. The body of the mask shall be a seamless, porous, non-woven fabric consisting of a mixture of polyester and rayon fibers, unified with a water-resistant resin. Testing shall be as specified in 4.4.

3.1.2.1.1 Color. The color of the body of the mask shall be blue approximately the Munsell Color Company's Color Sample Code 10B 6/10, glossy finished, except that the color need not be uniform in the ribbing and periphery areas. The inside area of the body of the mask shall be light blue approximating the Munsell Color Company's Color Sample Code 10B 8/4, except that the color need not be uniform throughout.

3.1.2.1.2 Porosity of mask fabric. The porosity shall be such that (a) the pressure differential across a sample shall not exceed 0.09 inches of water when a 17.5 square inch specimen from the center of the mask is exposed to an enclosed air stream of 50 liters per minute. The porosity of the mask material shall be tested as specified in 4.4.2.1 and (b) the filtering efficiency shall not be less than 95 percent when tested as specified in 4.4.2.2

3.1.2.2 Nosepiece. The nosepiece shall be fabricated from a single strip of malleable aluminum, rounded on both ends. The dimensions shall be 0.019 $\pm$ 0.002 inch thick, 3-1/2 $\pm$ 1/64 inches long and 3/16 $\pm$ 1/64 inch wide; shall be tested as specified in 4.4. The strips shall be malleable enough to be readily bent to, and maintain, the contours of the wearer's nose, so as to prevent the loss of expired air between the mask and the face in this area.

3.1.2.2.1 Color. The color of the nosepiece shall be the natural color of the metal, blue-green, or white.

3.1.2.3 Elastic band. The elastic band that is attached to the sides of the mask shall be fabricated from heat-resistant natural rubber or synthetic latex rubber. The length of the unstretched elastic band between attachment points shall be 15 inches $\pm$ 1 inch. The elastic band shall extend at least 3/8 inch beyond each staple. The width of the elastic band shall be not less than 1/16 inch nor more than 3/16 inch and shall be not less than 0.020 inch nor more than 0.030 inch thick. The elastic band shall not deteriorate when subjected to autoclaving and shall be tested as specified in 4.4.2.3.

3.1.2.3.1 Color. The color of the elastic band may be white, blue, or blue-green to match the body of the mask.

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3.1.2.4 Staples. The staples used to secure the elastic band to the sides of the body of the mask shall be fabricated from flat, galvanized low-carbon steel wire. The stapling wire shall have a thickness of 0.017 inch ± 0.003 inch.

3.1.2.4.1 Color. The staples shall be the natural color of the metal used.

3.2 Construction.

3.2.1 Type I and II mask. The mask shall consist of a filter mat encased and bonded in two soft layers of nonwoven fabric. The filter mat shall be constructed of the material specified in 3.1.1.2. The layers of non-woven fabric shall be constructed of the material specified in 3.1.1.1. The mask shall have a pouch or it shall be pleated to allow for a snug fit at the edges and without binding at the nose or mouth. When the mask has a pouch, the pouch shall be formed by a deep fold across the mask. The sides of the pouch shall be heat sealed. The peripheral margins of the masks shall be bound with the tape specified in 3.1.1.3. The tape along the horizontal or vertical margins shall be extended to serve as tie tapes. The four tie tapes shall be a minimum of 12 inches long from each side of the mask. When the tapes are tied the mask must conform to the contour of the wearer's nose and cheek. A nosepiece as specified in 3.1.1.4 shall be placed on the outside or inside the horizontal edge of the mask, as applicable. When the nosepiece is on the outside of the mask, it shall be secured so that it will not detach during use or when soaked with perspiration. When the mask is constructed with the metal strip inside the top horizontal edge, the word "INSIDE" shall be legibly stamped in ink on the middle of the top horizontal inside margin, except when the cover material on the inside of the mask is of a different color than the cover material on the outside.

3.2.1.1 Dimensions. The finished mask shall have the following dimensions:

Horizontal (Length--------6-3/8 to 7/34 inches.
Vertical (Width)----------3-1/8 to 4-1/4 inches.
Pouch, when applicable:
  Depth-----------1-1/2 inches ± 1/8 inch.
  Width (at narrowest point)------4-1/2 inches + 1/4 and -1/8 inch.
  Pouch starts ---1-1/2 inches ± 1/8 inch from top.
  Snout angle----30° to 37° from vertical margin.
  Maximum width of heat seal--------1/4 inch.
Pleats, when applicable:
  Opened width---------6-1/4 inches to 7-3/4 inches.
Tie Tapes--------A minimum of 12 inches long from each side of the mask.

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3.2.2 Type III mask. The body of the mask shall be molded or formed into a cup-shape, the periphery of which shall be contoured to fit over the bridge of the nose, under the eyes, along the cheeks, and under the chin. The cupping of the body shall be sufficiently deep so that the wearer's nose and mouth do not touch the mask. The portion of the cup over the nose and mouth shall be ribbed or corrugated transversely. The metal nose-piece shall be permanently cemented to the outside surface of the mask and approximately centered over the nose area with the ends not more than 1/2 inch from closest edge of mask. Each end of the elastic band shall be permanently attached to the outside surface of the mask with a metal staple. Each staple shall be attached 1/8 to 5/8 inch from edge of mask but not in the rib area. The elastic band and staple shall be assembled so that the elastic band does not break or pull out, and the staples do not pull out of the body of the mask when tested as specified in 4.4.2.3. The length of the unstretched elastic band, when measured between the points of attachment shall be 15 inches $\pm$ 1 inch. The body of the mask shall serve as a filter. The characteristic of the formed coated fabric shall be such that the body of the mask shall retain its shape under normal handling and use, but will permit adjustment and conformance to the contours of the face. The body of the mask shall not have an offensive odor.

3.2.2.1 Autoclaving of complete mask. The complete mask shall be capable of being sterilized in an autoclave without any deleterious effect, however a discoloration of the staples during autoclaving may occur. Complete mask shall be tested as specified in 4.4.2.4.

3.2.3 Performance characteristics.

3.2.3.1 Filtering efficiency. The filtering efficiency of the mask shall be not less than 95 percent when tested by any of the methods of 4.4.1.1, except 4.4.1.1.2 (Method-II). When tested by 4.4.1.1.2 (Method-II), the minimum filtering efficiency of the mask shall be 90 percent.

3.2.3.2 Air exchange pressure. (Ease of breathing). The air exchange pressure of the mask shall be a maximum of 5 millimeters (.197 inch) of H_2O, per square centimeter of sample tested. The air exchange pressure of the mask shall be tested as specified in 4.4.1.2.

3.2.3.2.1 Air exchange pressure alternate. The air resistance shall be 0.15 to 0.55 inches of water differential per 85 liters air flow per minute passing through the mask. The air resistance of the mask shall be tested as specified in 4.4.1.2.1.

3.2.3.3 Flammability. The flammability requirements of the mask shall be determined in accordance with Section IV of Commercial Standard 191-53 and the time of the flame spread shall be 4 seconds or more.

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3.2.3.3.1 Flammability alternate. The flammability requirements of the mask shall be determined in accordance with section 4.4.1.3.1. The time of the flame spread shall be 3 seconds or more.

3.2.3.4 Certificate of Dermatological reactions. The contractor shall submit to the procuring activity, a copy of the original certification, that the lots or batches offered to the Government were composed of the same formula and manufactured identically as the production lot or batch originally tested for dermatological reactions to the impregnated fabric, when tested as specified in 4.4.3 or 4.4.3.1.

3.3 Workmanship. The mask shall be free of cuts, tears or any other imperfections or defects that will detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examination and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificates of quality supplied by the manufacturer of the component or material may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105, except when otherwise specified herein.

4.3 Examination of the end item. The masks shall be examined for visual defects. The sample unit for this examination shall be one mask. The inspection level shall be S-4 of MIL-STD-105. The acceptable quality levels shall be 1.5 for major defects and 6.5 for minor defects, expressed in defects per hundred units.

4.3.1 Classification of defects. Examination shall be conducted in accordance with, but not restricted to the following classification of defects.

4.3.1.1 Type I, II and III mask.

Categories and defects	
Major	
101	Any holes, cuts or tears in body of mask.
102	Mask without pleats or pouch (Type I and II only).
103	Body of mask not contoured to face, not molded or not formed into cup shape (Type III only).
104	Ribs omitted or do not transverse the body of the mask (Type III only).
105	Nosepiece omitted or not securely attached.
106	Nosepiece not malleable to the extent specified.
107	Mask constructed without four tie-strings (Type I and II only).
108	Elastic band omitted or not securely attached (Type III only).
109	Filter mat encased but not bonded to two soft layers of non-woven fabric (Type I only).
110	Mask not bound on all four sides (Type I and II only).
111	Mask having offensive odor.
Minor	
201	Visible spots or specks of foreign matter on body of mask.
202	Mask not clean (not free of lint or loose fibers).
203	Wrinkles on body of mask (Type III only).
204	The word "INSIDE" not marked on inside top horizontal margin, when specified (Type I and II only).
205	Nosepiece not placed as specified.
206	Tie strings having thread ends not trimmed at the end (Type I and II only).

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4.3.2 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level Shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table I. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements, and as indicated in table II. The unit of product for testing shall be one mask. The sample size shall be determined in accordance with special inspection level S-2 of MIL-STD-105, except for the characteristic "filtering efficiency" which shall be determined in accordance with special inspection level S-1. The number of test failures shall be zero.

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Table I - Tests

	Characteristic	Requirement Paragraph	Test Method
Fabric Type III only	Identification of polyester and rayon	3.1.2.1	1/
	Fabric construction	3.1.2.1	Visual
	Color	3.1.2.1.1	Visual
	Porosity	3.1.2.1.2	4.4.2.1
	Filtering efficiency	3.1.2.1.2	4.4.2.2
	Flammability	3.2.3.3	4.4.1.3
	Dermatology	3.2.3.4	4.4.3
Facing material Type I and II only	Identification of material (Rayon)	3.1.1.1	1/
	Fabric construction	3.1.1.1	Visual
	Color	3.1.1.1	Visual
	Dermatology	3.2.3.4	4.4.3
Filter media Type I and II only	Identification of filter media	3.1.1.2	1/
	Fiber diameter (Average)	3.1.1.2	1/
Mask Type I and II only	Width of tie tapes	3.1.1.3	Visual
	Dimensions	3.2.1.1	Visual
	Filtering efficiency	3.2.3.1	4.4.1.1
	Air exchange pressure	3.2.3.2	4.4.1.2
	Flammability	3.2.3.3	4.4.1.3
Mask Type III only	Elastic & Staple assembly	3.2.2	4.4.2.3
	Length of elastic band between staples	3.2.2	Visual
	Autoclaving	3.2.2.1	4.4.2.4
Nosepiece Type III only	Metal; identification of aluminum	3.1.2.2	1/
	Dimensions	3.1.2.2	Visual
	Color	3.1.2.2.1	Visual
Nosepiece Type I and II only	Identification of material (metal)	3.1.1.4	1/
	Dimensions	3.1.1.4	Visual
	Thickness	3.1.1.4	Visual (gauze)
	Color	3.1.1.4.1	Visual
Tie Tapes Type I and II only	Identification of material	3.1.1.3	1/
	Color	3.1.1.3.1	Visual

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Table II - Tests (cont'd)

	Characteristic	Requirement Paragraph	Test Method
Elastic band Type III only	Material Identification	3.1.2.3	1/
	Width	3.1.2.3	Visual
	Thickness	3.1.2.3	Visual (gauge)
	Autoclaving	3.1.2.3	4.4.2.4
	Color	3.1.2.3.1	Visual
Staples Type III only	Metal, Identification Low-carbon galvanized steel wire	3.1.2.4	1/
	Thickness, gauge	3.1.2.4	Visual
	Color	3.1.2.4.1	Visual

1/ A certificate of conformance shall be submitted by the supplier and will be acceptable for the stated requirement.

4.4.1 Type I and Type II mask.

4.4.1.1 Filtering efficiency. The filtering efficiency of the type II mask shall be determined in accordance with 4.4.1.1.1 or 4.4.1.1.2 or by method Greene and Vesley (J. Bacterial. Vol. 83, No. 3, P-663, or a comparable "in vitro" method. (see 4.4.4).

4.4.1.1.1 Method - I. Cultures --- Staphylococcus aureus, Utah strain 15 (see 6.4) cultures are transferred at least 3 times in nutrient broth (BBL) before preparation of the test suspension. Each transfer is incubated at 37° ± 2°C. for 24 to 28 hours, constantly agitated throughout the incubation period on a mechanical shaker.

Aerosol filtration of bacteria. The efficiency of each mask is tested with the apparatus shown in Figures 1 and 2 in which an Andersen sampler (see Fig. 1) is utilized to collect and count bacteria on nutrient agar (BBL) in special petri dishes.

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The bacterial culture is first diluted to a given turbidity as may be measured by an adequate optical instrument (i.e. Klett-Summerson or Coleman Jr. spectrophotometers) such that when the optically adjusted suspension is diluted approximately 1/3500, the nutrient agar control plates will yield a total colony count of 2200 ± 500 colonies. The diluted culture suspension is fed into a nebulizer from a syringe driven by a constant drive mechanism (see figure 2). The nebulizer is operated at 15 pounds per square inch pressure to produce an aerosol in the aerosol chamber. The aerosol is drawn through the apparatus at a flow rate of 28.3 liters per minute from one minute to establish control values. The air flow is measured with a Roger Gilmont Instruments (RGI) turned off and the chamber and sampler washed with 28.3 liters of air. The control plates are removed from the sampler and replaced immediately with fresh nutrient agar plates. Thereafter, circular specimens, 3 inches in diameter, of each surgical mask to be tested are interposed, in turn, at the point where the aerosol chamber is attached to the Andersen sampler. The bacterial aerosol is again drawn through the apparatus, identical to the control run, except for the presence of the mask material. All plates are incubated at 37° ± 2°C. for 48 hours and the colonies counted.

The filtration efficiency is reported in "Percent Efficiency".

$$\text{Percent Efficiency} = \frac{C-F}{C} \times 100$$

Where, C = the average number of colonies on control plates.
F = number of colonies on test plates (mask material interposed).

4.4.1.1.2 Method-II . The apparatus used in method II is shown in figure 3. An aerosol of bacteria is produced by a combination of air pressure at points "I" and "A" and vacuum at points F, G, and H. The flow rates are 1,900 milliliters per minute at the positive pressure end and 3,320 milliliters per minute at the vacuum end of the apparatus. The positive pressure is directed through a Vaponephrin nebulizer (point B) containing approximately two milliliters of a suspension of 5,000 viable cells of Staphylococcus aureus 209 per milliliter of 0.1 percent peptone water. This suspension is pipetted into

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the nebulizer. Such a suspension is prepared from frozen viable cells which shall be counted prior to test. The aerosol is directed into a mixing chamber (point C) in which the larger droplets of the bacteria laden aerosol are removed. The aerosol is drawn by a controlled vacuum applied at the vacuum end, through the face mask secured in the mask holder (point D) and are trapped on the millipore filter membrane which is 47 mm in diameter having a porosity of 0.8 u (micron) and placed at Point E. Alternate samples are taken with and without masks in the system and a time period of five minutes is required for each sample tested. Each filter membrane is placed onto a nutrient agar plate and incubated for twenty-four hours at 37°C. At the end of the incubation period the colonies shall be counted using a dissecting microscope or a Quebec Colony Counter. The filtering efficiency of the masks is calculated as follows:

$$\text{Efficiency in } \% = \frac{(\text{Count without mask}) - \text{Count with mask} \times 100}{\text{count without mask}}$$

4.4.1.2 Air exchange pressure. A device which measures the pressure differential required to draw air through a measured surface area at a constant air flow rate is used to measure the air exchange pressure of various materials (see figure 4). Water filled manometers, (m_1, and m_2) are used to measure the pressure differential. A Roger Gilmont Instrument (R.G.I.) flowmeter or a flowmeter of comparable precision is used for measurement of the air flow. An electric vacuum pump draws air through the apparatus and a needle valve is used to adjust the flow rate. The test material is placed across the 2.5 cm diameter orifice and clamped into place so that the test material will be in line and across the flow of air. The pump is started and the flow of air adjusted to 8 liters per minute. Manometers m_1 + m_2 are read and recorded. This procedure is carried out on five different areas of the mask and the readings averaged. The air exchange pressure is calculated as follows:

$$\Delta P = \frac{\bar{X}_{m_2} - \bar{X}_{m_1}}{4.9}$$

Where: $\bar{X}\ m_1$ = mm water pressure, Manometer No. 1 average of 5 test areas, low pressure side of test material.

$\bar{X}\ m_2$ = mm water pressure, Manometer No. 2-average of 5 test areas, high pressure side of test material.

4.9 = cm^2 area of test material.

Δ P = Pressure differential per cm^2 of test material expressed as mm of water.

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NOTE: In lieu of two manometers, a Differential Inclined Draft Gauge, graduated in tenths and calibrated in inches of water may be used. Pressure differential may be reported in inches of water.

4.4.1.2.1 Air exchange pressure alternate. This method is applicable to the determination of the air resistance of a Disposable Mask. The resistance to a fixed flow of air by the mask is measured by a differential draft gauge.

Equipment.

Fischer & Porter Flowrator No. 10A1027A with hose clamp connections. Similar to F&P Meter No. B4-27-10/27 with stainless steel float. Maximum capacity 100 liters air per min. calibrated in percent of maximum flow. Calibrations, 10 to 100, graduated in intervals of 10. (Figure 5 - No. 1)

Dwyer Differential Draft Gauge graduated 0.10 to 0 to 1.0 in tenths and calibrated in inches water. (Figure 5 - No. 2).

Face Mask Holder and Holding Frame has been designed to be affixed to the Dwyer Differential Draft Gauge (see Figure 5)(3 & 4)

Procedure

Connect apparatus as shown in Figure 5.

The F&P Flowrator shall be plumb. The Dwyer Inclined Gauge shall be level and plumb.

Allow sufficient plastic or rubber tubing in the connection between equipment No. 1 and No. 3 (Figure 5) to facilitate the insertion and removal of test mask samples. Approximately 30 inches of tubing should be sufficient. Mount the Dwyer Inclined Draft Gauge directly above the mask holder and holding frame. Connect mask holder to reservoirs of Draft Gauge by flexible tubing as shown in the diagram.

Remove mask holder from holding frame by relieving tension on mask holder. Use adjusting hand wheel on left hand member of mask holder. Insert face mask in right hand member of mask holder so that flutes of the mask are downward. Fit right hand member into mask opening which has been placed into left member of holding frame. Insure that a perfect seal has been obtained between face mask and both halves of the mask holder. Insert mask holder into mask holding frame maintaining hand pressure against the face mask by the two halves of the holding frame. Tighten adjusting wheel of the holder as far as possible to insure a complete seal around the entire face mask in order to prevent air leaks.

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just the scale of the Dwyer Inclined Gauge to zero. Introduce air from air pply adjusting volume of air at 85 on F&P Flowrator (this is equivalent to liters air per minute).

ad the mask resistance (inches water differential) from the Dwyer Inclined uge.

4.4.1.3 Flammability. The basic materials of the mask shall be subcted to the required tests of paragraph 4 of Commercial Standard CS 191-53.

4.4.1.3.1 Flammability alternate. This method is also applicable to the termination of fire retardancy of non-woven fabrics.

ie prepared specimen is held at an angle of 45 degrees, a standardized ame is held to its lower edge and the time required for flaming to proed up the fabric a distance of five inches is recorded. The test should e conducted in a draft free chamber.

quipment

lammability Tester - Model 7633, or equivalent with five specimen holders ach consisting of two 1/16" thick matched metal plates with clamps mounted l the side. (NOTE: These plates have an opening in the center 1 1/2" ide and at least 6" long. The plates are slotted and loosely pinned for lignment. The pins are placed so that they may be used for a thread guide " above the point at which the flame is applied.)

. standard tank of butane with 2 pound capacity equipped with a sensitive control valve.

n oven capable of maintaining a temperature of 221°F.

Sample Preparation

Cut five samples 2" x 6" with the 6" dimension in the machine or long fiber direction.

Place the samples in the specimen holders (exposing 5 1/4" of the sample) and dry in an oven for 1/2 hour at 221°F.

If the sample is damaged by drying for 1/2 hour at 221° F, dry in a desiccator (over calcium sulfate) for 24 hours.

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Place the specimen holders with sample in a desiccator over anhydrous calcium chloride and allow to cool to ambient temperature (minimum time - 45 minutes).

Procedure

Step 1. Adjust the fuel valve so that the flame coming from the gas jet is 5/8" long measured from the tip of the flame to the opening in the gas nozzle.

Step 2. Mount the specimen on the rack at a 45° angle within 45 seconds.

Step 3. Attach the weight to the end of the cotton thread and place the thread across the face of the specimen so that it is 5 inches above the point where the flame will be applied.

Step 4. Suspend the weight above the stop watch so that when the thread is broken, the weight will drop and strike the watch in such a way as to stop the timing action.

Step 5. Apply the flame perpendicular to the sample and at the same time start the timing cycle.

Step 6. Hold the flame in this position until the fabric ignites, or, for 20 seconds. No ignition after 20 seconds is called "Did Not Ignite" (DNI).

Step 7. The burning time is the elapsed time between application of the flame and the burning through of the thread.

Report the average of five tests.

4.4.2 Type III mask.

4.4.2.1 Porosity test. The porosity of the mask material shall be tested across a circular sample, encompassing the ribbed part of the mask, having an area of 17.5 square inches. This area shall be exposed, perpendicular to the air flow, in a closed air stream having a continuous flow rate of 50 liters of air per minute. The pressure differential from the inside of the mask to the outside, shall be measured on a standard inclined manometer.

4.4.2.2 Filtering efficiency. The filtering efficiency of the mask shall be determined by the method of Greene and Vesley (J. Bacterial Vol. 83, No. 3, P-663, March 1962), or a comparable "in vitro" method using the Anderson Sampler (4.4.4).

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4.4.2.3 Elastic and staple assembly. The elastic and staple assembly shall be tested by grasping the elastic band midway between the staples and extending it twice to a distance of 23 inches from the staples. The elastic shall be completely relaxed between each extension.

4.4.2.4 Autoclaving. The complete mask shall be subjected to autoclaving at 250° Fahrenheit, 15 pounds pressure for not less than 20 minutes.

4.4.3 Dermatology test. Dermatological reaction tests shall be performed by the methods described in the Code of Federal Regulations, Title 21 - Food and Drugs, Part 147 to End, Revised as of January 1, 1971, Paragraphs 191.10 and 191.11 page 248, with values from table for skin reactions of 1 or less, or that method as approved by the contracting officer.

4.4.3.1 Dermatology test alternate. Dermal Toxicity as described in the publication" Association of Food and Drug Officials of the United States - Appraisal of the Safety of Chemicals in Foods, Drugs, and Cosmetics - Published 1959, page 46 J.H. Draize, section 3 for Eye Mucosa. This procedure is modified as follows:

1) Do not wash the eye
2) Instill extract into one eye of each of six mature albino rabbits on each of three consecutive days.

Observation shall indicate that no substance which possessed irritative potential of significance will leach from the mask.

4.4.4 When alternate comparable tests, not listed in this specification, are selected to determine filtering efficiency, the test procedure shall be reviewed for approval by the Directorate of Medical Materiel, ATTN: ATT, of the Defense Personnel Support Center, Philadelphia, Pa. 19101.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing fifty type II or type III masks or three hundred type I masks, as specified constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package.

5.1.2.1.1 Type I. Fifty masks shall be packaged in a dispenser type box that permits one-at-a-time removal. Closure shall be adequate to prevent accidental opening. Six dispenser type boxes each containing fifty masks each shall be further packaged in a snug fitting box constructed in accordance with PPP-B-636, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.2.1.2 Type II. Fifty type II masks shall be packaged in a dispenser type box as specified in 5.1.2.1.

5.1.2.1.3 Type III. Fifty type III masks shall be packaged in a snug fitting box constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.2.2 Intermediate package. Six units containing type II masks or twelve units containing type III masks shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, class domestic. Closure shall be adequate to prevent accidental opening. An intermediate package shall not be required for type III masks.

5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2)

5.2.1 Level A. Four units of type I masks, twenty four units of type II masks or forty eight units of type II masks shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Masks shall be packed as specified in 5.2.1 except that the exterior container shall conform to PPP-B-636, class domestic and the closure shall conform to method II as specified in the box specifications.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The masks covered by this specification are intended for use by Military Medical Services.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing. (see 5.1 and 5.2).

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog.

NSN 6515-00-782-2621 Mask, Surgical, Disposable, Pouch, Blue or Green, Type II
NSN 6515-00-982-7493 Mask, Surgical, Disposable, Molded, Blue, Type III
NSN 6515-01-003-3142 Mask, Surgical, Disposable, Pleated, Blue or Green, Type I.

6.4 Cultures of Staphylococcus aureus, Utah strain 15, are available from:

Dr. Paul S. Nicholes
Dept, of Microbiology
University of Utah Medical Center
Salt Lake City, Utah 84112

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1930
is

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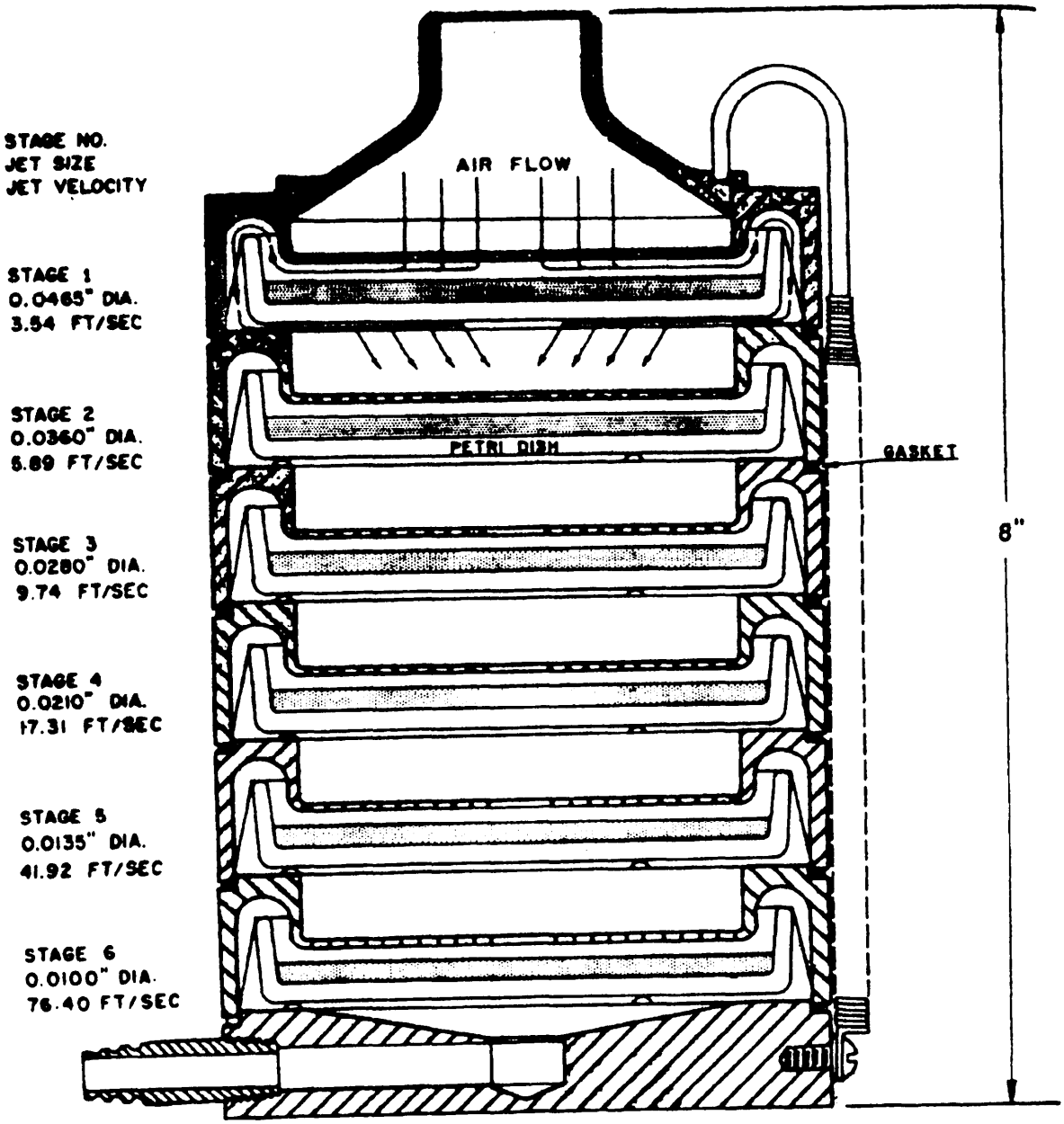


Figure 1

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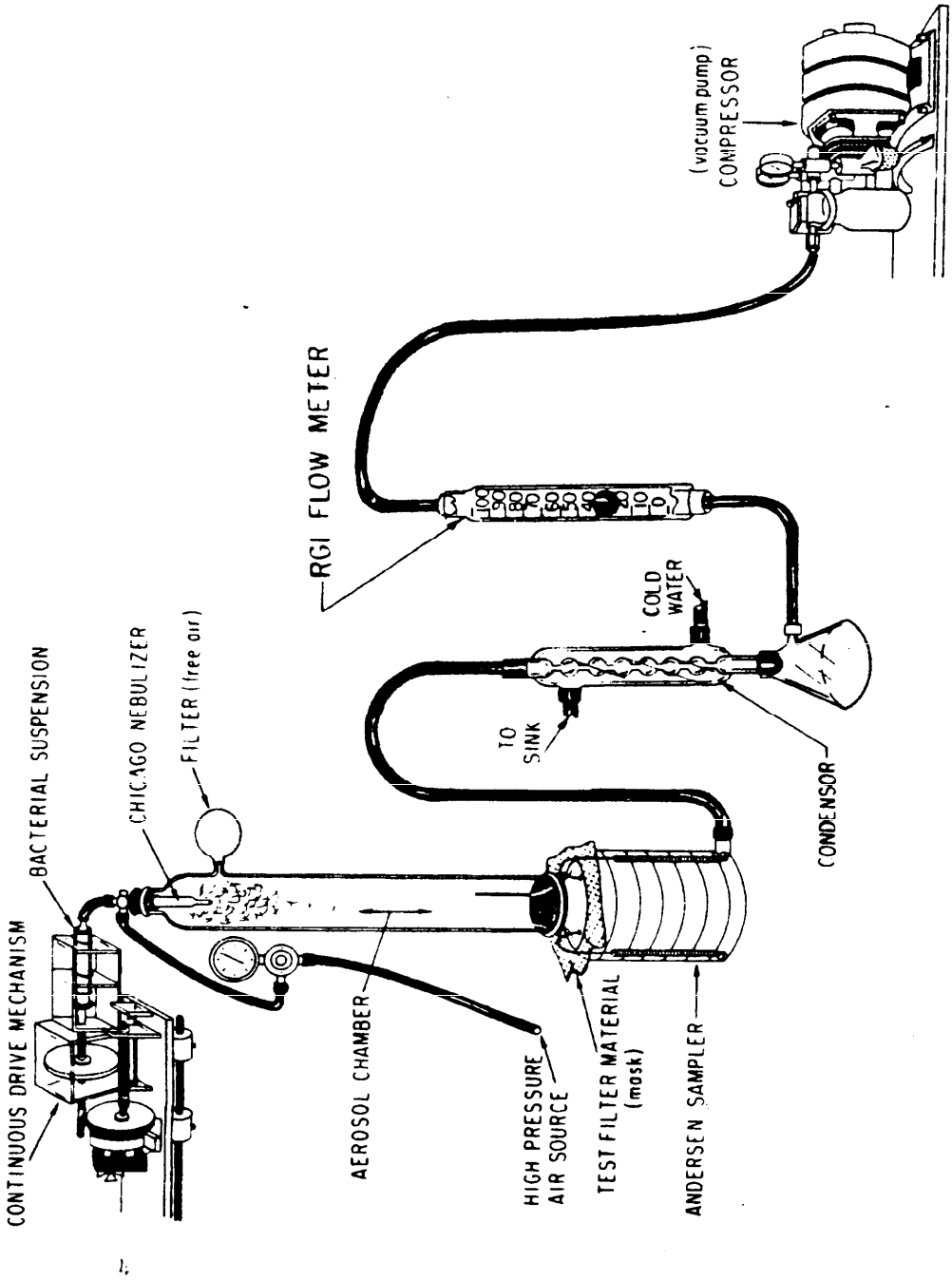
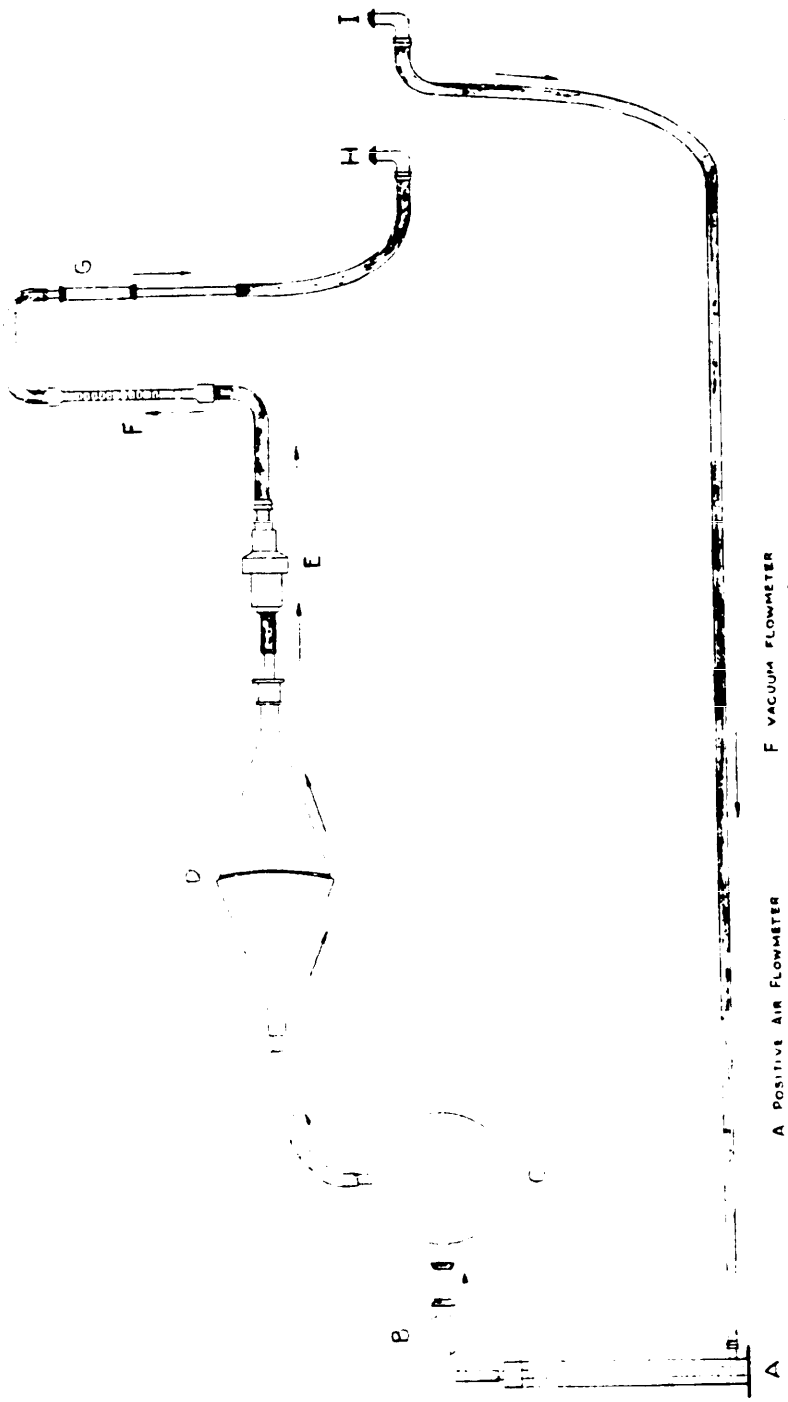


FIGURE 2

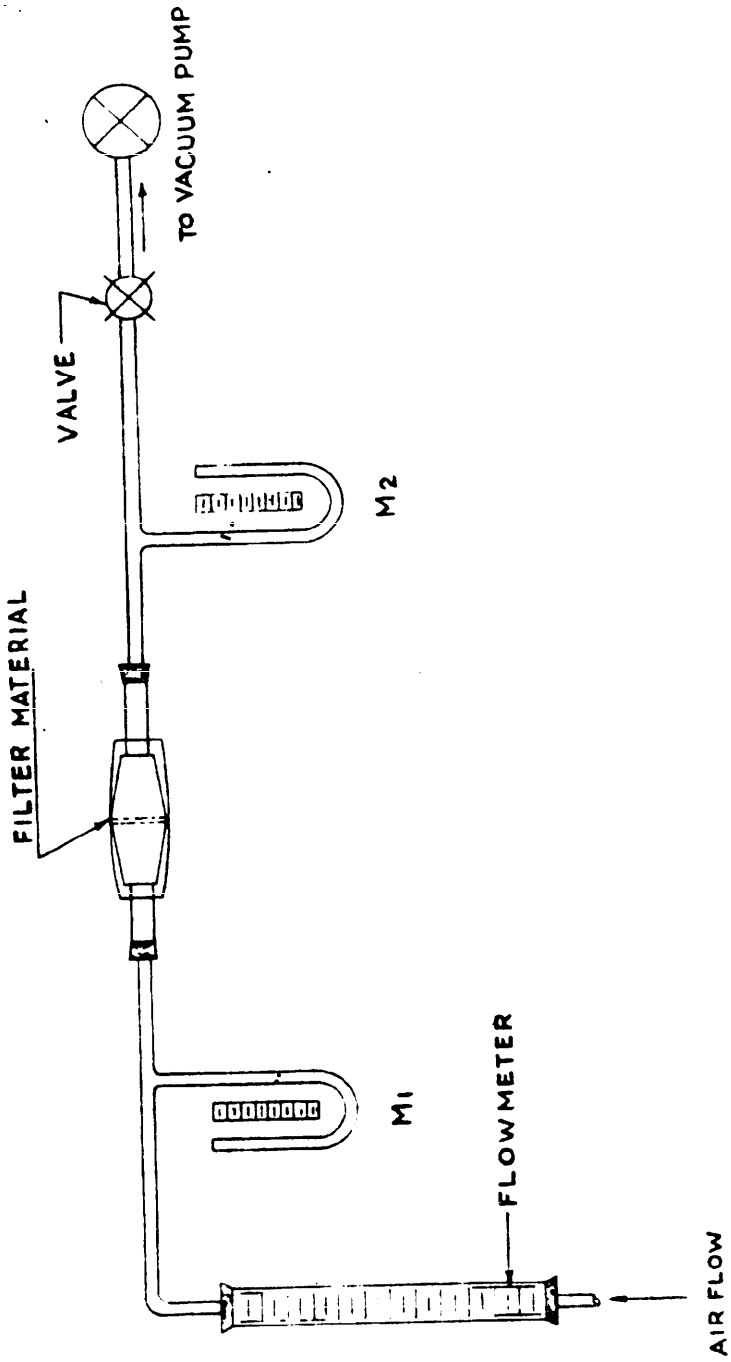
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BACTERIAL AEROSAL FILTRATION APPARATUS

FIGURE 3

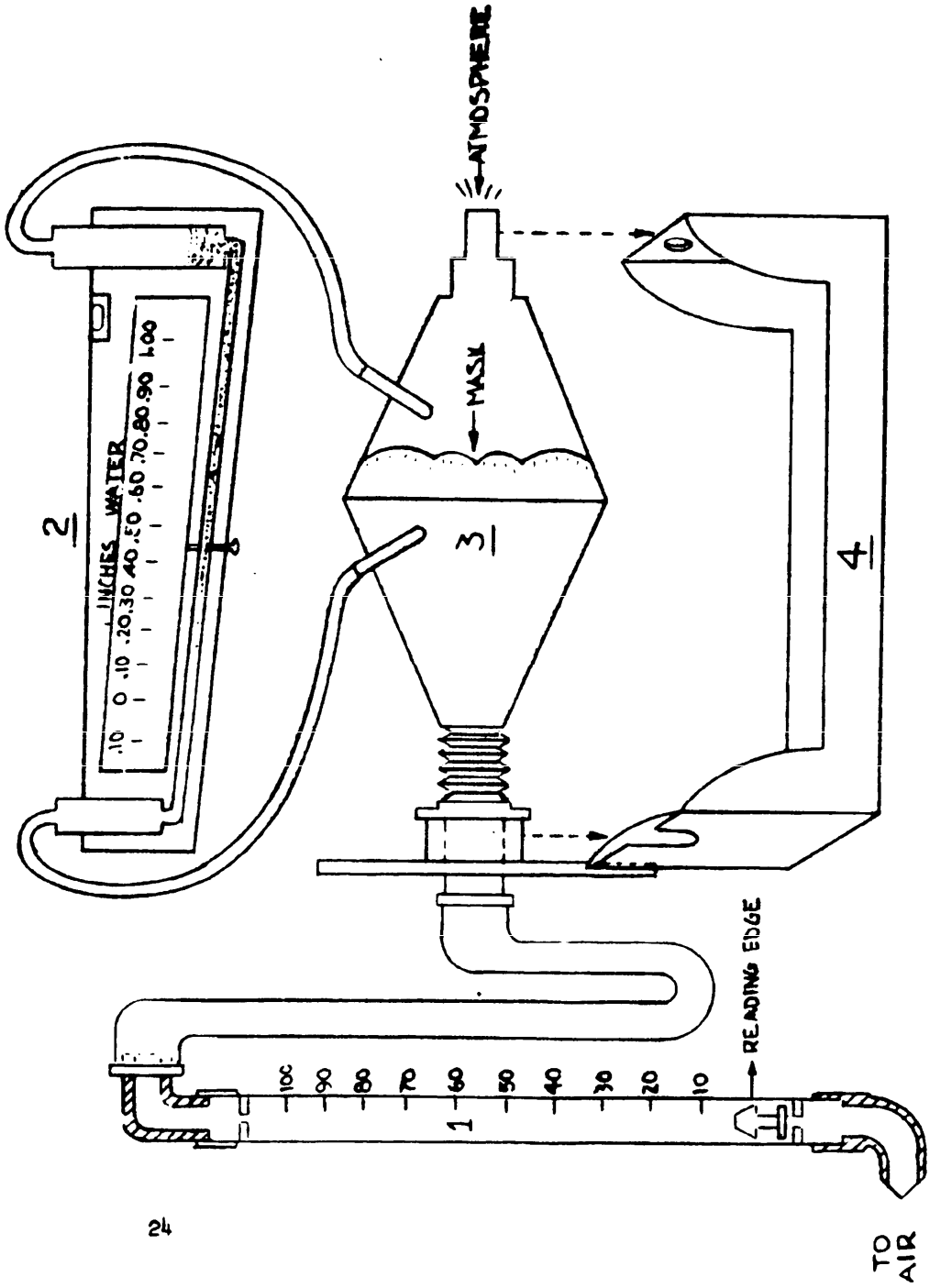
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AIR PRESSURE-FLOW MEASURING APPARATUS

Figure 4

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TEST EQUIPMENT FOR DETERMINING AIR RESISTANCE OF DISPOSABLE FACE MASK.

FIGURE 5

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

MIL-M-36954C MASK, SURGICAL, DISPOSABLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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